YafaNER at KnowledgeGraphEval 2026: AraBERTv2-based Approach to Cross-Domain Adaptation for Arabic NER

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Abstract

Cross-domain Arabic Named Entity Recognition (NER) remains challenging because models trained on one domain can lose performance when applied to unseen domains with different vocabulary, writing styles, and entity distributions. This paper addresses this challenge, and presents Yafa's Team submission (ranked 7th) to Subtask 1 of the KnowledgeGraphEval 2026 Shared Task. Our system leverages an AraBERTv2-based model for nested Arabic NER to approach it. The proposed approach combines task-adaptive (TAPT) pretraining on unlabeled Konooz data, R-Drop fine-tuning on Wojood, strong data augmentation, teacher-student knowledge distillation, span-level confidence and entropy filtering, and an entity-specific confidence-aware weighted-logit ensemble. Three teacher models generate soft labels for Konooz, and three distilled student models are combined in the final ensemble to produce the final predictions. Wojood contains approximately 448K tokens and 72K entity mentions across 21 entity types, while Konooz contains data from 10 domains for cross-domain evaluation. The final system achieves a Micro F1 of 71.33% on the official Konooz test set, improving over the non-adapted AraBERTv02 baseline of 61.00% by 10.33 points. The results show that combining domain adaptation, filtered knowledge transfer, and entity-specific ensembling can improve cross-domain Arabic NER performance.

1 Introduction

Named Entity Recognition (NER) is a fundamental information extraction task that identifies mentions of people, organizations, locations, dates, and other semantic classes in text. NER plays a vital role in many applications, such as word sense disambiguation (Jarrar et al., 2023), knowledge graph construction, and relation extraction (Jarrar et al., 2022; Aljabari et al., 2025b). Arabic NER remains particularly challenging due to rich morphology and complex orthographic variations. Domain shift across domains and dialects adds a further layer of difficulty: a model trained on a specific domain often suffers significant degradation when applied to another (Hamad et al., 2025a). Moreover, annotated resources remain limited and often differ in tagsets and coverage (Hamed, 2019; Ali et al., 2020). These challenges are especially pronounced in cross-domain NER, where models trained on one domain often suffer substantial performance degradation when applied to unseen domains, reflecting differences in entity definitions, boundaries, and lexical context (Hamad et al., 2025a; Aljabari et al., 2025a). This makes robust domain-adaptive Arabic NER a necessary step toward reliable knowledge extraction across heterogeneous text sources.

This paper outlines the Yafa's Team (YafaNER is ranked 7th on the leaderboard) submission to Subtask 1 of the KnowledgeGraphEval 2026 Shared Task (Aljabari et al., 2026) at the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026). Where we tackle the cross-domain adaptation for Arabic NER. This shared task focuses on recognizing and classifying named entities in Arabic text while evaluating the ability of systems to generalize across unseen domains. It is based on the AdaptNER benchmark, where models are trained on one or more source domains and evaluated on different target domains to measure cross-domain robustness.

This paper makes the following three contributions. First, we present an enhanced AraBERTv2-based system for cross-domain, nested Arabic named entity recognition. This system integrates task-adaptive pretraining (TAPT), R-Drop fine-tuning, entity-preserving strong data augmentation, teacher-student knowledge distillation, and ensemble learning. Second, we propose a span-level confidence and entropy filtering strategy to select reliable teacher-predicted entity spans from the unlabeled Konooz corpus prior to knowledge distil-

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1244–1249
October 28-29, 2026 ©2026 Association for Computational Linguistics

lation. This reduces the impact of noisy supervision during student training. Third, we develop an entity-specific, confidence-aware weighted logits ensemble. The ensemble weights are derived from per-entity validation F1 scores using a temperature-controlled softmax function. This enables each entity type to benefit from the model that performs best for that entity type. Our final system achieved a Micro F1 score of 71.33% on the official Konooz test set.

The paper is organized as follows: Section 2 presents prior and recent research on Arabic AdaptNER. Section 3 provides an overview of the datasets used. Sections 4 and 5 describe the proposed systems and experimental setup. Section 6 presents and discusses our experimental results. Finally, Section 7 concludes the paper and suggests ideas for future work.

2 Related Work

Arabic NER has progressed from rule-based and feature-engineered systems to neural sequence models and transformer encoders. Early Arabic systems relied on handcrafted lexical, morphological, and gazetteer features (Shaalan and Raza, 2007; Benajiba et al., 2007; Abdul-Hamid and Darwish, 2010; Abdallah et al., 2012; Mohit et al., 2012). BiLSTM-CRF and related neural models improved robustness by learning distributed representations directly from data (Lample et al., 2016; Khalifa and Shaalan, 2019). The shift to pretrained transformers further advanced Arabic NER, with AraBERT, ARBERT, and MARBERT providing stronger contextual representations and better handling of morphology and orthographic variation (Antoun et al., 2020; Abdul-Mageed et al., 2021).

Arabic corpora often differ substantially in entity inventories and boundary conventions. *Wojood* highlights this issue by showing that entity definitions vary even within Arabic, across datasets such as ANERCorp, AQMAR, and OntoNotes, making direct transfer unreliable (Jarrar et al., 2022; Aljabari et al., 2025a). These divergences are especially harmful in cross-domain settings, where models trained under one genre or annotation guideline must generalize to unseen domains with different vocabulary, style, and entity distributions.

Wojood demonstrated that a single corpus can cover MSA and dialectal Arabic across multiple domains while supporting nested annotations and a richer tagset than prior Arabic resources (Jarrar et al., 2022). $Wojood_{fine}$ extends this scheme with finer-grained subtypes, while $Wojood^{Gaza}$ and *Konooz* further broaden domain and dialect coverage (Liqreina et al., 2023; Jarrar et al., 2024; Hamad et al., 2025a).

Cross-domain Arabic NER therefore remains an open problem at the intersection of representation learning and dataset harmonization. This gap motivates **KnowledgeGraphEval 2026 Shared Task Subtask 1 (AdaptNER)**, which requires systems that do not merely memorize a single corpus but generalize reliably across heterogeneous Arabic domains.

3 Datasets and Stats

This work relied on two main Arabic datasets, *Wojood* (Jarrar et al., 2022) and *Konooz* (Hamad et al., 2025b), as well as an augmented version of the training data generated using Strong Data Augmentation; each dataset was used at a different stage of the proposed system.

3.1 Wojood

Wojood is a corpus for Arabic nested NER. It contains approximately 448K tokens and 72K entity mentions annotated with 21 entity types. Wojood is the official labeled dataset used in AdaptNER subtask competition to train and evaluate models *for nested named entity recognition*. The dataset consists of three main files, which are corresponding to the training, validation, and test sets. Each line in the Wojood files contains a token followed by one or more BIO tags, depending on the number of overlapping entities associated with that token.

3.2 Augmented Wojood Training Set

Since the original training data was limited compared to the diversity of domains present in the test data, an augmented version of the Wojood training set was created using Strong Data Augmentation.

The augmented data retains the same Wojood data format and the same tagging scheme (Nested BIO) to ensure compatibility with the model. Thus, new new sentence versions were generated using various data augmentation techniques while maintaining tag consistency as much as possible. Specifically, for approximately 30% of the training sentences, one of three entity-preserving operations: O-only word dropout, O-only word masking, or O-only adjacent word swap was randomly applied to O-tagged tokens , while entity-bearing tokens (B-/I- tags) were left unchanged.

3.3 Konooz

Konooz is the official test set for the competition. It is an unlabeled dataset used to evaluate the models' ability to generalize to new domains (cross-domain generalization). The dataset consists of 10 files, each representing a different domain. As for the Konooz files, each line contains a token followed by 21 columns representing entity types, all of which have a value of **O** because this is an unlabeled dataset.

Overall, the following note applies to the three datasets: Each pair of sentences within the three datasets is separated by a blank line. This format defines sentence boundaries during training and evaluation.

4 System Description

We used AraBERTv2-based models to label the tokens found in the development and test sets, known respectively, as "dev-konooz" and "blinded-test-data". Here, we provide a brief description of the base models, followed by a detailed description of the variant models used for submission on Codabench. Code and released resources are available at `https://github.com/ohamed/Arabic-NER`.

4.1 AraBERT and AraBERTv2

AraBERT (Antoun et al., 2020) is an Arabic-specific, pretrained transformer model that extends the Google's BERT architecture. It was pre-trained on corpora of Arabic news articles comprising tens of millions of sentences.

AraBERTv2 is the updated version of AraBERT model presented by Antoun et al. (2020) *AraBERTv2 is an advanced, transformer-based pre-trained language model built specifically for the Arabic language*. AraBERTv2 was trained on a massive 77GB dataset (200 million Arabic sentences or 8.6 billion words) of Arabic text.

4.2 AraBERTv2 Ensemble

Our proposed system is based on AraBERTv2 and uses the BertNestedTagger architecture to identify nested named entities in Arabic text. The model performs token-level classification with parallel BIO classification heads, which allow multiple entity labels to be assigned to the same token, thereby supporting nested entity recognition. It combines TAPT, R-Drop regularization, teacher-student knowledge distillation, span-level confidence, and entropy filtering. Our system was developed as follows:

First, TAPT was performed on the unlabeled Konooz corpus using only its raw, unlabeled text via masked language modeling. No gold labels were used at any stage prior to the final prediction. Then, the resulting model was fine-tuned on the Wojood dataset using R-Drop. Three teacher models were trained with random seeds 1, 42, and 2026. Strong data augmentation was applied to the third teacher model to increase prediction diversity. These three teacher models were then combined using a weighted logits ensemble to generate soft labels for the unlabeled Konooz corpus. The teacher-predicted entity spans were then filtered using span-level confidence and entropy thresholds to retain only the most reliable entity spans.

Next, three student models were trained using teacher-student knowledge distillation. Each student model learned jointly from the manually annotated Wojood dataset and the filtered Konooz teacher outputs through soft supervision.

The final system is an ensemble of the three distilled student models. During inference, their predictions are combined using an entity-specific, confidence-aware weighted logit ensemble. The ensemble weights are derived from per-entity validation F1 scores using a temperature-controlled softmax function. A temperature of 0.03 produced the best validation performance and was thus employed in the final submission. Figure 1 shows the general sequence of stages followed to label Konooz files using our proposed AraBERTv2-based ensemble system.

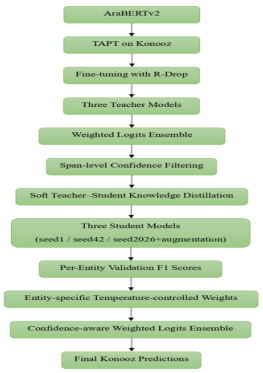


Figure 1: The development of AraBERTv2 ensemble.

5 Experimental Setup

This system does not use prompting, retrieval, or external knowledge sources. It relies exclusively

on AraBERTv2, the Wojood dataset, and the unlabeled Konooz corpus. All phases of the project were carried out using the Google Colab platform. The NER models were trained using AdamW with a learning rate of 1×10^{-5}, a batch size of 8, a maximum sequence length of 512, and a dropout rate of 0.1. R-Drop used $\alpha = 1.0$, while knowledge distillation used a temperature of $T = 2.0$ and a weight of 0.5. These values were empirically selected based on preliminary experimentation with the validation set.

5.1 Preprocessing

Before implementing the proposed training stages, a series of data preprocessing steps were performed to prepare the datasets for the requirements of each stage. These steps fall into four categories: (i) preparing Konooz for task-adaptive pretraining; (ii) tokenization and sequence preparation; (iii) preparing Konooz for NER prediction; and (iv) creating the augmented Wojood training set.

5.2 Final System Configuration

As shown in Figure 1, we conducted our experiments iteratively until we reached the final system configuration. The final system is an AraBERTv2 ensemble (AraBERTv2-E), which is an entity-specific, confidence-aware weighted logits ensemble. The adopted hyperparameter values are listed in Table 2 in Appendix A.1, along with the rationale behind key choices (e.g., the ensemble temperature).

6 Results

Table 2 presents the performance of the final AraBERTv2 ensemble (AraBERTv2-E) on the official Konooz test set compared with the baseline model. AraBERTv2-E improves the Micro F1 from 0.61 to 0.7133, an improvement of 10.33 percentage points. The final system obtains a Weighted F1 of 0.7069 and a Macro F1 of 0.6091. The difference between the Macro and Weighted F1 scores shows that performance is not uniform across entity types.

Model	Micro F1	Macro F1	Weighted F1	Precision	Recall
BERT w/o adaptation (AraBERTv02)	0.61	–	–	–	–
AraBERTv2-E w adaptation	**0.7133**	0.6091	0.7069	0.7325	0.6952

Table 1: Performance comparison of the evaluated models.

Table 3 in Appendix A.2 shows how each system component contributes to the Dev-Konooz and official Konooz test sets.

6.1 Error Analysis

An error analysis of the models forming AraBERTv2-E, based on the labeled Wojood test set, reveals difficulties in distinguishing some contextual expressions from named entities. For example, in "اجتماع نابلس", the gold annotation labels only "نابلس" as GPE, while all three models forming the final ensemble incorrectly identify "اجتماع نابلس" as an EVENT. Another difficulty appears in nested entity recognition. In "انتخابات المجلس الإسلامي الأعلى", the gold annotation contains an EVENT spanning the full expression and an ORG nested within it. One of the student models identifies the inner ORG but misses the enclosing EVENT, while another correctly identifies both entities. These examples show that contextual ambiguity and overlapping entity structures remain challenging for the models used in the final system.

6.2 Limitations

One limitation of this work is that the Konooz test set was unlabeled during system development, so direct error analysis on the target-domain predictions was not possible. In addition, the experiments were conducted using AraBERTv2-based models, and other Arabic pretrained language models were not evaluated.

7 Conclusion

This paper presented our approach for the KnowledgeGraphEval 2026 Shared Task, Subtask 1 on cross-domain Arabic NER. Our AraBERTv2-based system combines task-adaptive pretraining, R-Drop fine-tuning, strong data augmentation, teacher-student knowledge distillation with span-level confidence and entropy filtering, and an entity-specific confidence-aware weighted logits ensemble. It achieved a Micro F1 score of 71.33% on the official Konooz test set. Future work will investigate other Arabic pretrained language models, small LLMs, improved confidence estimation and knowledge transfer methods, and more effective use of unlabeled data for cross-domain adaptation.

Acknowledgments

We thank Palestine Technical University - Kadoorie and Forschungszentrum Jülich for their support.

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A Additional Experimental Details

A.1 Hyperparameter Configuration

Table 2 lists the main hyperparameters of the final ensemble system.

The per-entity ensemble weights were derived from the Wojood validation F1 score. The single

Parameter	Value
Input Models	3 Student Models
Ensemble Level	Logits
Weighting Strategy	Entity-Specific Weights
Weight Source	Per-Entity Validation F1
Weight Normalization	Softmax
Temperature (T)	0.03
Output	Final Prediction Files

Table 2: Main hyperparameters used for AraBERTv2-E.

ensemble-temperature value was selected by comparing four candidates (0.03/0.05/0.07/0.10) via the official Konooz leaderboard, with the score spread (0.7128–0.7133) indicating low sensitivity to this choice.

A.2 Incremental Ablation Results

To isolate the contribution of each component, we report in Table 3 incremental results on Dev-Konooz used for comparative evaluation during system development, alongside results on the official Konooz test set used for the shared-task submission. Results obtained on the two evaluation sets are not directly comparable.

Stage	Evaluation Set	Micro F1	Δ F1
Baseline	Dev- Konooz	0.6272	—
TAPT	Dev- Konooz	0.6870	+0.0598
TAPT + R-Drop	Dev- Konooz	0.6961	+0.0091
TAPT + R-Drop+ Strong Augmentation	Dev- Konooz	0.7083	+0.0122
Fixed Weighted Logits Ensemble	Dev- Konooz	0.7155	+0.0072
Fixed Weighted Logits Ensemble	Official Konooz	0.7102	—
Entity-specific Ensemble	Official Konooz	0.7128	+0.0026
+ Confidence Weighting	Official Konooz	0.7132	+0.0004
+ Temperature = 0.03	Official Konooz	0.7133	+0.0001

Table 3: Incremental contribution of each system component on the Dev-Konooz and official Konooz test sets.